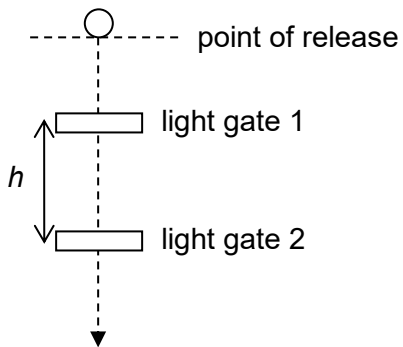


4

To determine the acceleration of free fall, a steel ball is dropped above two light gates as shown.

The ball passes light gate 1 and 2 at times t_1 and t_2 after release.



[Turn over

What is the acceleration of free fall?							
A	$\frac{2h}{t_2 - t_1}$	B	$\frac{2h}{(t_2 - t_1)^2}$	C	$\frac{2h}{t_2^2 - t_1^2}$	D	$\frac{2h}{\left(\frac{t_2 + t_1}{2}\right)^2}$

Ans: C

Let distance to light gate 1 from point of release be x

$$x = ut + \frac{1}{2}at^2 = \frac{1}{2}gt_1^2$$

$$x + h = \frac{1}{2}gt_2^2$$

$$h = \frac{1}{2}gt_2^2 - \frac{1}{2}gt_1^2$$

$$g = \frac{2h}{t_2^2 - t_1^2}$$

5	Three boxes are pushed along a smooth surface by a force F as shown below. What is the
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